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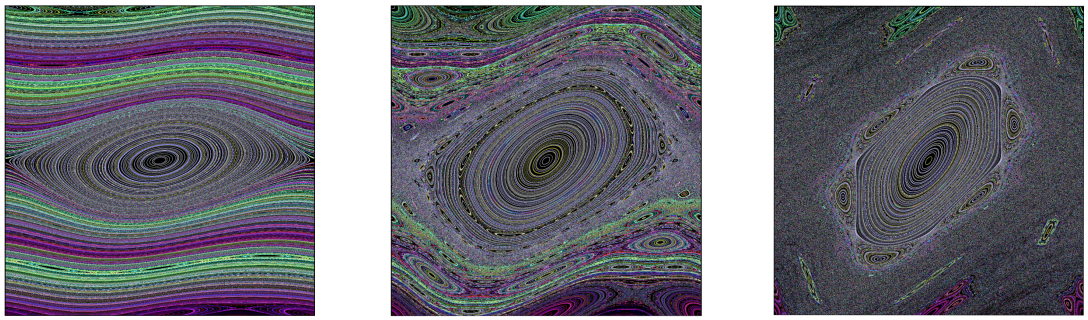
Dynamics of the kicked rotor



This report is an English translation of the original Italian version.

Abstract

This project describes and analyses the dynamics of the **kicked rotor**. We explain what a kicked rotor is and how its dynamics can be described by the **standard map**. We use a numerical model to compute its trajectories in phase space. This allows us to identify chaotic behaviour in certain regions of phase space and quantify it by numerically estimating the **Lyapunov exponents**.

1 Introduction

The study of dynamical systems that exhibit chaotic behaviour is an area of great scientific interest. Examples include the formation of thunderstorms, the motion of fluids, and even the motion of three bodies governed by Newton's law of universal gravitation. These systems are deterministic in the sense that, given an initial condition, their behaviour can in principle be determined exactly in both the future and the past. They are also chaotic. Two initial conditions that are so close as to be indistinguishable with the instruments available can lead to trajectories that rapidly and inexorably diverge. This is the familiar *butterfly effect*.

In this project we study a very simple chaotic system. It consists of a rod that is free to rotate about a point and receives a *kick* at every time interval ΔT. A kick is an instantaneous force that imparts a finite amount of momentum to the system. Despite its apparent simplicity, the *kicked rotor* can exhibit chaotic behaviour for certain initial conditions. We quantify this behaviour by numerically computing the *Lyapunov exponents*, defined in Section 4. We thus show how such a simple system can exhibit behaviour that is extremely difficult to predict analytically without a numerical model.

All programs used in this project are written in Python using *NumPy* and *Matplotlib* and are available here.

2 Equations of motion of the kicked rotor and the standard map

In this section we derive the equations of motion of the kicked rotor using the Hamiltonian formulation of mechanics. The Newtonian or Lagrangian formulation would yield the same equations. There are, however, reasons why the Hamiltonian formulation is particularly useful and interesting in this case.

First, the Hamiltonian formulation applies to all systems, including non-mechanical ones, that are described by equations of the form

$$\frac{d\mathbf{x}}{dt} = S\nabla_{\mathbf{x}}\mathscr{H},$$

where $\mathbf{x} \in \mathbb{R}^{2n}$ represents a state of the system, $\mathscr{H} : \mathbb{R} \times \Omega \to \mathbb{R}$ is its Hamiltonian (Ω is an open subset of $\mathbb{R}^{2n}$), and $S \in M(2n, \mathbb{R})$ is the **symplectic matrix**

$$S := \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

Systems of this kind are called **Hamiltonian dynamical systems**. If a dynamical system is Hamiltonian and its Hamiltonian satisfies certain regularity assumptions, we can use theorems of Hamiltonian mechanics, such as Noether's theorem or Liouville's theorem, to simplify the analysis considerably.

Second, the kicked rotor can also be studied from a quantum-mechanical perspective, for which the Hamiltonian formulation is convenient. A quantum treatment of the system is beyond the scope of this project.

We now determine the Hamiltonian of the kicked rotor. For simplicity, suppose the rotor is a rod of moment of inertia I pivoted at one end. We choose orthonormal Cartesian axes $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ and use the Hamiltonian coordinates (θ, p), where θ is the angle the rod makes with the y axis and $p = I\dot{\theta}$ is the momentum conjugate to θ. In an inertial frame at rest relative to the rod, the Hamiltonian is simply the mechanical energy of the system. In the absence of kicks it is given by

$$\mathscr{H} = \frac{p^2}{2I}. \tag{1}$$

We now introduce the kicks. Suppose that at $t = T$ a constant force acts on the system in the $-\hat{\mathbf{y}}$ direction and imparts a finite amount of momentum. Since the force is constant and instantaneous, it must take the form

$$\mathbf{F}(t) = -K\delta(t - T)\hat{\mathbf{y}},$$

where δ is the Dirac delta. Since $\mathbf{F}$ is constant in space, we can associate it with a potential $V(\mathbf{r}) = -K\delta(t - T)y$. We can therefore assign the rod a potential energy due to the kick,

$$U = K\ell\cos(\theta)\delta(t - T),$$

where ℓ is the length of the rod. For simplicity we absorb ℓ into K, so that $U = K\cos(\theta)\delta(t - T)$.

We want the kicks to occur whenever $t = kT$ with $k \in \mathbb{Z}$, so we replace $\delta(t - T)$ with

$$\sum_{k\in\mathbb{Z}} \delta(\frac{t}{T} - k),$$

which is a Dirac delta that is "activated" whenever $t = kT$ instead of only at $t = T$. The Hamiltonian of the *kicked*

rotor is therefore

$$\mathscr{H} = \frac{p^2}{2I} + K\cos(\theta)\sum_{k\in\mathbb{Z}}\delta(\frac{t}{T}-k), \tag{2}$$

from which Hamilton's equations follow directly

$$\begin{cases} \dfrac{d\theta}{dt} = \dfrac{\partial\mathscr{H}}{\partial p} = \dfrac{p}{I} \\ \dfrac{dp}{dt} = -\dfrac{\partial\mathscr{H}}{\partial\theta} = K\sin(\theta)\displaystyle\sum_{k\in\mathbb{Z}}\delta(\frac{t}{T}-k) \end{cases}. \tag{3}$$

This system of equations presents some difficulties. It is non-autonomous, and the presence of $\sin(\theta)$ and the Dirac delta in the second equation makes it nonlinear. As expected, when $t \neq kT$ the rod rotates freely, with θ increasing linearly in time and p constant. Since the behaviour between two kicks is straightforward, instead of solving (3) for $(\theta(t), p(t))$, we can restrict our attention to the values of θ and p immediately before the nth kick, denoted by θ_n and p_n. If we assume that θ is constant on $[nT-\epsilon, nT+\epsilon]$ for sufficiently small ϵ, integrating the second equation in (3) from $nT-\epsilon$ to $nT+\epsilon$ gives

$$p_{n+1} - p_n = \int_{nT-\epsilon}^{nT+\epsilon} K\sin(\theta)\delta(\frac{t}{nT}-k)dt = KT\sin(\theta_n),$$

where we have used the fact that p is constant between kicks. Integrating the first equation in (3) from $nT-\epsilon$ to $nT+T-\epsilon$ gives

$$\theta_{n+1} - \theta_n = \frac{Tp_{n+1}}{I}.$$

Combining these results gives

$$\begin{cases} \theta_{n+1} = \theta_n + \frac{T}{I}p_{n+1} \\ p_{n+1} = p_n + KT\sin(\theta_n) \end{cases}. \tag{4}$$

We can simplify the system further by introducing the *dimensionless momentum* $P = \frac{T}{I}p$ and the *chaos parameter*, which we also denote by K, through $K \to \frac{T^2}{I}K$. Equation (4) then becomes

$$\begin{cases} \theta_{n+1} = \theta_n + P_{n+1} \\ P_{n+1} = P_n + K\sin(\theta_n) \end{cases}. \tag{5}$$

We have obtained a relation between (θ_{n+1}, P_{n+1}) and (θ_n, P_n). Given an initial value (θ_0, P_0), we can use this relation to determine the evolution of the system much more easily than with (3). The equations in (5) are known as the **standard map** or the **Chirikov map**.

Before computing kicked-rotor trajectories numerically with the standard map, we make some qualitative observations. As expected, the standard map is nonlinear. We might expect $\theta = \pi$ to be a stable equilibrium of the system. We can therefore shift the angle by $\theta \to \theta - \pi$ so that the equilibrium is at $\theta = 0$ and the angle ranges from $-\pi/2$ to $\pi/2$ instead of from 0 to 2π. Expanding $\sin(\theta)$ to first order about $\theta = 0$, we obtain from (5)

$$\begin{cases} \theta_{n+1} = \theta_n + P_{n+1} \\ P_{n+1} = P_n - K\theta_n \end{cases}.$$

In matrix form, this is

$$\begin{pmatrix} \theta_{n+1} \\ P_{n+1} \end{pmatrix} = \begin{pmatrix} 1-K & 1 \\ -K & 1 \end{pmatrix}\begin{pmatrix} \theta_n \\ P_n \end{pmatrix}. \tag{6}$$

For values of θ near π, we can therefore approximate the standard map by a linear map. We can substitute into (6) the value of (θ_n, P_n) obtained from the $(n-1)$th iteration of the map. Repeating this substitution until we reach (θ_0, P_0) gives

$$\begin{pmatrix} \theta_n \\ P_n \end{pmatrix} = \begin{pmatrix} 1-K & 1 \\ -K & 1 \end{pmatrix}^n\begin{pmatrix} \theta_0 \\ P_0 \end{pmatrix}. \tag{7}$$

This is a kind of "analytical solution" of the standard map, although in this form it is difficult to see how $\theta(n)$ and $P(n)$ behave. If, however, we diagonalise over $\mathbb{C}$ the matrix

$$M := \begin{pmatrix} 1-K & 1 \\ -K & 1 \end{pmatrix},$$

that is, if we find two values $\lambda_1, \lambda_2 \in \mathbb{C}$ and a matrix $N \in GL(2,\mathbb{C})$ such that $M = NDN^{-1}$, where $D = \mathrm{diag}(\lambda_1, \lambda_2)$, we can write

$$M^n = N\begin{pmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{pmatrix}N^{-1}. \tag{8}$$

Carrying out the calculation, (6) simplifies to

$$\begin{cases} \theta_n = \frac{\lambda_1\lambda_2}{\lambda_1-\lambda_2}\left[(\lambda_2^{n-1}-\lambda_1^{n-1})\theta_0 + (\frac{\lambda_2^{n-1}}{\lambda_1} - \frac{\lambda_1^{n-1}}{\lambda_2})P_0\right] \\ P_n = \frac{\lambda_1\lambda_2}{\lambda_1-\lambda_2}\left[(\lambda_1^n-\lambda_2^n)\theta_0 + (\frac{\lambda_1^n}{\lambda_2} - \frac{\lambda_2^n}{\lambda_1})P_0\right] \end{cases}.$$

Here $\lambda_1 = -\frac{K+\sqrt{K^2-4K}}{2} + 1$ and $\lambda_2 = -\frac{K-\sqrt{K^2-4K}}{2} + 1$. To understand the meaning of these expressions, we assume for simplicity that $P_0 = 0$. This gives

$$\theta_n = \lambda_1\lambda_2\frac{\lambda_2^{n-1}-\lambda_1^{n-1}}{\lambda_1-\lambda_2}\theta_0. \tag{9}$$

As long as $K < 4$, λ_1 and λ_2 are complex conjugates, so $\lambda_1\lambda_2$ is real and $\lambda_1 - \lambda_2$ is imaginary. If we write $\lambda_1 = Re^{i\omega}$, $\lambda_2 = Re^{-i\omega}$ and $\lambda_1 - \lambda_2 = 2i/C$, where $C, R \in \mathbb{R}$, equation (9) becomes

$$\begin{aligned} \theta_n &= CR^{1+n}\theta_0\frac{e^{i\omega(n-1)} - e^{-i\omega(n-1)}}{2i} \\ &= CR^{1+n}\theta_0\sin(\omega(n-1)). \end{aligned} \tag{10}$$

A similar calculation gives

$$P_n = CR^{2+n}\theta_0\sin(\omega n). \tag{11}$$

Equations (10) and (11) are particularly interesting because they let us predict the system's trajectories[1] when it starts at a point (θ_0, P_0) close to $(0, 0)$. In particular, we have the following cases.

- When $R < 1$, the trajectory spirals into $(0, 0)$.
- When $R = 1$, the trajectory is an ellipse centred at $(0, 0)$.
- When $R > 1$, the trajectory diverges and our analysis is no longer valid.

Calculating R explicitly as a function of K gives $R = 1$ for $0 \leq K \leq 4$ and $R > 1$ for $K > 4$. Thus, when we compute the trajectories numerically, we expect to find ellipses around $(0, 0)$ at least for $K \leq 4$.[2] We call $(0, 0)$ and the other points around which orbits form in phase space **stable points**.

Before computing trajectories, we make one final observation. From the original definitions of θ and P, the phase space is $[0, 2\pi) \times \mathbb{R} \subset \mathbb{R}^2$. The standard map (5) shows that if (θ_n, P_n) is a trajectory, then $(\theta_n, P_n + 2k\pi)$, where $k \in \mathbb{Z}$, must also be a trajectory, since θ_n is 2π-periodic. In other words, we can reduce the phase space to the subset $[0, 2\pi)^2$, since every trajectory with $|P_n| > 2\pi$ is equivalent to one with $|P_n| < 2\pi$. Moreover, the states $(\theta, 0)$ and $(\theta, 2\pi)$ are equivalent, as are $(0, P)$ and $(2\pi, P)$. The phase space therefore has the topology of a *torus*. This observation will be useful when we compute the trajectories numerically.

This symmetry has a physical interpretation. For a given trajectory (θ_n, P_n), the trajectory $(\theta_n, P_n + 2\pi)$ is equivalent to (θ_n, P_n) in the sense that the rod has the same orientation immediately before each kick. In the second case, the rod makes one extra revolution between successive kicks.

3 Numerical computation of trajectories in phase space

As discussed above, (5) is much more convenient than (3) for computing the evolution of the system numerically. We proceed as follows.

- Fix a value of K.
- Divide $[0, 2\pi) \times [-\pi, \pi)$ into a grid of M^2 points and assign each grid point a colour.
- For each grid point $(\theta_i(0), P_j(0))$, simulate the system using the standard map for a total of N iterations.
- Generate a square image with a resolution of 1024×1024 pixels. Colour every pixel visited by trajectory ij using the colour assigned to its starting grid point.
- If more than one colour is assigned to a pixel, take their average.[3]

The images were generated with $M = 50$ and $N = 1000$ for different values of K. The images immediately show that, as predicted, approximately elliptical orbits form around the centre of phase space, the point $(\pi, 0)$.

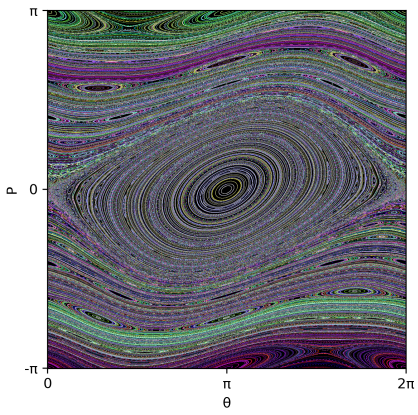


Figure 1: Phase space for $K = 0.6$

We also see that when K exceeds a certain threshold, chaotic regions begin to appear. In these regions the trajectories cease to form clearly visible orbits. These regions grow as K increases, as can be seen by comparing Figures 1 and 3.

In the next section, we compute the Lyapunov exponents to verify this qualitative observation.

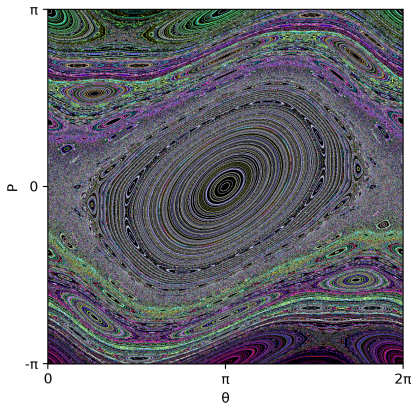


Figure 2: Phase space for $K = 0.9$

Besides the stable point at $(\pi, 0)$, there are other stable points. Their number and positions appear to depend on K.

In the previous section we showed that for $K \leq 4$ there is a central stable point around which the trajectories follow ellipses. Although our argument applied only for

[1] Here trajectories refer to the points (θ_n, P_n), as opposed to the "full" trajectories $(\theta(t), P(t))$, which can in general intersect and are less convenient to work with.

[2] All calculations here use the change of coordinates $\theta \to \theta - \pi$, so the point $(0, 0)$ here is the point $(\pi, 0)$ in the original coordinates.

[3] The average is taken over the triplets of numbers representing the RGB colours.

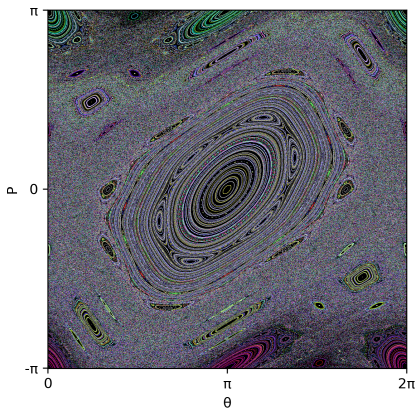


Figure 3: Phase space for $K = 1.2$

$K \leq 4$, we can conjecture that for $K > 4$ the stable point ceases to exist.

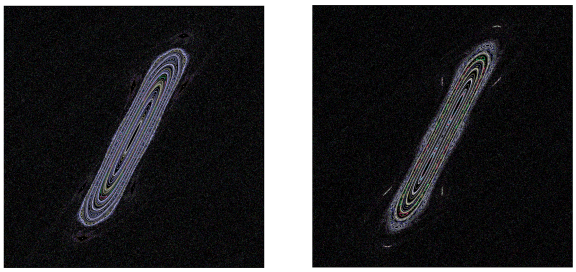


Figure 4: Phase space for $K = 3.9$ (left) and $K = 4.1$ (right)

Figure 4 shows that this is what happens.

4 Lyapunov exponents of the standard map

In the previous section we observed qualitatively that the phase space of the *kicked rotor* contains both chaotic and regular regions. We now want to quantify chaos near a given configuration. For this we use **Lyapunov exponents**.

Consider an m-dimensional dynamical system, where m is the dimension of phase space. The system described by the standard map, for example, is two-dimensional. Suppose the system is described by a differentiable map $F : \mathbb{R}^m \to \mathbb{R}^m$. Given a point x_0 and a nearby point x in phase space, we want to quantify how quickly the trajectory starting at x_0 separates from the one starting at x. If the distance between the trajectories grows exponentially in time, we call the region around x_0 chaotic. At $n = 0$, the distance between x and x_0 is $\delta(0) =\| x - x_0 \|$. After n iterations of F it is

$$\delta(n) =\| F^n(x) - F^n(x_0) \|,$$

where F^n denotes the nth iterate of F. Since F is differentiable, to first order we have

$$\delta(n) \approx\| DF^n(x_0)(x - x_0) \|,$$

where DF^n is the Jacobian of F^n, given by

$$DF^n(x_0) := \prod_{k=0}^{n-1} DF(x_k).$$

When DF is constant, this is simply the nth power of $DF(x_0)$. If we assume that δ grows by a factor λ at each iteration, the geometric mean of these factors over the evolution of the system is

$$\lambda = \lim_{n\to\infty} \left(\frac{\delta(n)}{\delta(0)}\right)^{\frac{1}{n}} = \lim_{n\to\infty} \left(\frac{\| DF^n(x_0)(x - x_0) \|}{\| x - x_0 \|}\right)^{\frac{1}{n}}.$$

If we set $\mathbf{v} := \frac{x-x_0}{\|x-x_0\|}$ and take the logarithm of λ, we obtain the *Lyapunov exponent* of F at x_0 in the direction $\mathbf{v}$

$$\lambda_L(x_0, \mathbf{v}) = \lim_{n\to\infty} \frac{1}{n} \log(\| DF^n(x_0)\mathbf{v} \|). \tag{12}$$

This definition immediately gives the first-order evolution of δ for a point x close to x_0

$$\delta = \delta(0)e^{n\lambda_L(x_0,\mathbf{v})}. \tag{13}$$

For a fixed x_0, it is natural to ask how λ_L changes with $\mathbf{v}$. One can show that there is a basis $\mathbf{e}_1, ..., \mathbf{e}_m$ of phase space and a set of values $\lambda_1 \leq \lambda_2 \leq ... \leq \lambda_m$ with the following properties.

- If $\mathbf{v} = v\mathbf{e}_k$, then $\lambda_L(x_0, \mathbf{v}) = \lambda_k$.
- If $\mathbf{v}$ is a linear combination of $\mathbf{e}_{n_1}, ...\mathbf{e}_{n_k}$, then $\lambda_L(x_0, \mathbf{v}) = \max(\lambda_{n_1}, ...\lambda_{n_k})$.

For more details, see [1] and [2]. This is particularly useful because, for a randomly chosen direction $\mathbf{v}$, $\lambda_L(x_0, \mathbf{v})$ is very likely to be $\max(\lambda_L(x_0, \mathbf{e}_1), ..., \lambda_L(x_0, \mathbf{e}_m))$. From now on we consider only this last value, which we denote by $\lambda_L(x_0)$ for simplicity.

Unfortunately, for the standard map we cannot use the approximation that DF is constant, and computing the product of the n matrices $DF(x_n)$ analytically is rather laborious. There is also another difficulty. The definition above assumes that phase space is a subset of $\mathbb{R}^m$, whereas for the standard map it is a torus. This leads to some interesting results. For example, [3] shows that for the standard map $\delta(n) \leq kn^2$ as $n \to \infty$, which implies $\lambda_L = O(n^{-1} \log n)$ as $n \to \infty$. This still leaves room to compute Lyapunov exponents for the "initial" evolution of the system, since, as we will see, λ_L does not immediately go to 0.

With this in mind, we compute the values of λ_L numerically as follows.

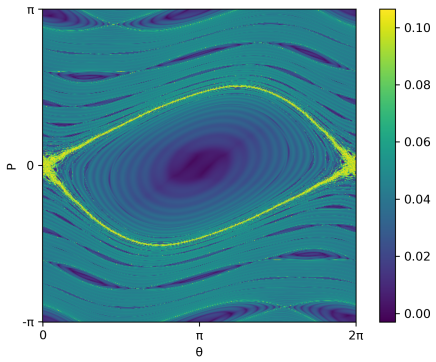


Figure 5: Lyapunov exponents for $K = 0.6$

- Define a function that takes an initial point x_0 and a vector $\mathbf{v}$, computes the distance $\delta(n)$ between the trajectories starting at x_0 and at $x_0 + \mathbf{v}$, and returns

$$\lambda = \frac{1}{n} \log \left(\frac{\delta(n)}{\delta(0)} \right),$$

  where $\delta(0) = \| \mathbf{v} \|$. We denote this function by `computeLyapunov(x0, v, n)`.
- Divide the phase space $[0, 2\pi) \times (-\pi, \pi]$ into an $M \times M$ grid.
- For each initial condition on the grid, choose two orthogonal vectors $\mathbf{v}$ of length dx and evaluate `computeLyapunov(x0, v, n)` for each. Denote the larger of the two values by λ.

We thus assign a value of λ to each initial condition on the grid. The results below use $M = 500$, $n = 100$ and $dx = 0.0001$.

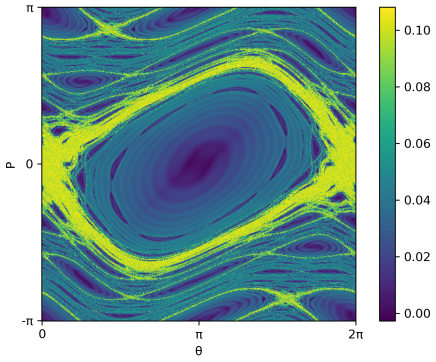


Figure 6: Lyapunov exponents for $K = 0.9$

The Lyapunov exponents allow us to locate the chaotic regions more accurately than we could by inspecting the trajectories in phase space.

In the central non-chaotic region, spirals appear to form around $(\pi, 0)$. It is possible that only some of the elliptical trajectories around the central point are actually stable, so that points starting off these trajectories may eventually decay onto the stable ones.

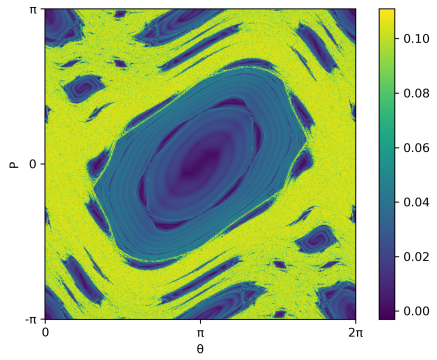


Figure 7: Lyapunov exponents for $K = 1.2$

I do not know whether there is a way to verify this behaviour analytically, but I suspect it is due to the nonlinearity of the standard map. In any case, the images suggest that the Lyapunov exponents tend to oscillate around all the stable points, producing patterns that almost resemble diffraction patterns. I do not think this is a computational error, since the same patterns are visible when the simulation is run at higher resolution, for example with $M = 1000$, $N = 500$ and $dx = 0.00001$.

5 Conclusion

Using Lyapunov exponents, we quantified the chaotic behaviour of the *kicked rotor* and identified the boundary between chaotic and non-chaotic regions more precisely. We also predicted analytically the behaviour of the stable point at $(\pi, 0)$, finding that it ceases to exist for $K > 4$. Many questions remain open. The first is whether the points we have called *stable points*, perhaps imprecisely, can be determined analytically. I initially hoped there might be a simple way to do this, perhaps by relating their positions to the zeros of a polynomial in $\mathbb{C}$ or something similar. I tried several approaches, but none worked. More advanced mathematics, beyond the scope of this project, is probably needed.

References

[1] J.M. Strelcyn G. Benettin, L. Galgani. Kolmogorov entropy and numerical exponents. *Physical review A*, 1976.

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